

Sandhya Nunemunthala

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PROFESSIONAL SUMMARY:

- **10+ Years** of experience in designing, developing, and maintaining **AWS (Amazon Web Services), GCP (Google Cloud Platform)** and **Microsoft Azure**
- Certified in both AWS and GCP
 - ⇒ **AWS Solution Architect Professional** – [Certification Link](#)
 - ⇒ **Google Cloud Certified Professional Data Engineer** – [Certification Link](#)
- Designed and developed POC Solutions for **Gen AI** Data Pipeline using **AWS Cloud** and **GCP Cloud**.
- Designed and developed multi-tenant data applications supporting **Data Warehouse, Data Lake** and **Data Mart** in **AWS** and **GCP Cloud**
 - ⇒ In **AWS Cloud** I have used services like: **Redshift, S3, RDS (Oracle, MySQL, MSSQL), Glue (Jobs, Connections, Notebooks, Crawlers & Catalog), EMR Cluster, Ec2, Step Function & Airflow, SNS Topic & SQS, Lambda (Functions & Layers), IAM (Roles & Policies), etc.,**
 - ⇒ In **GCP Cloud** I have used services like: **BigQuery, Cloud Storage, Cloud SQL (Oracle, MySQL), Data Flow, Data Proc, Compute Engine, Cloud Composer, Pubsub, Cloud Function, IAM (Roles & Service Accounts), ect.,**
- Hands-on experience designing and implementing **Generative AI pipelines** and **LLM-based solutions** across **AWS** and **GCP** cloud ecosystems to support **chatbots, agentic workflows, and context-aware automation**.
- Designed and developed **Proof-of-Concepts (POCs)** for **Gen AI** applications using **Amazon Bedrock, SageMaker JumpStart, Titan Embeddings, and Vertex AI** with **Gemini models** to explore use cases in customer service, search augmentation, and document intelligence.
- Created intelligent agents and workflow automations using **Bedrock Agents, Amazon Lex, Comprehend, and AWS Step Functions** to deliver context-aware, goal-driven task orchestration across cloud-native services.
- Designed and developed highly scalable data applications for Machine Learning, Data Science and Generative AI Solutions to support real-time and batch use case in **AWS** and **GCP Cloud**
 - ⇒ In **AWS Cloud** I have used services like: **Glue, Lambda, OpenSearch, SageMaker, S3, DynamoDB,**
 - ⇒ In **GCP Cloud** I have used services like: **Cloud Run, Cloud Function, AlloyDB, Vertex AI, Document AI, Gemini, BigQuery**
- Designed and developed **Model Monitoring** solutions to validate model performance and **model Metrix** in **AWS Cloud**
- Designed and developed feature store for **Machine Learning** and **Data Science** use cases in **AWS Cloud**
- Designed and developed real-time and batch reports and Dashboards in both **AWS** and **GCP** cloud

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⇒ In AWS Cloud I have used services like: **S3, Lambda, Glue, SNS, Cloud Watch Event StepFunction and QuickSight**

⇒ In GCP Cloud I have used services like: **Cloud Storage, Cloud Function, Pubsub, Cloud Scheduler, Workflow, Cloud Composer, BigQuery and Power BI**

- Extensive work experience in Implemented **continuous integration and continuous deployment (CI/CD) through** Bit Bucket and Jenkins using **UCD (UrbanCode Deploy)** also worked extensively using Gitlab and **Terraform**.
- Extensive experience in writing **Infrastructure as Code (IaaS)** in **Terraform, AWS CloudFormation**
- Designed and developed purpose-built layers using **AWS** services like **RDS (Oracle, MySQL), Redshift, DMS (Data Migration Service), S3, Cloud Watch alarms,**
- Developed complex mappings and loaded data from various sources into the **Data Warehouse, using different transformations/stages like Joiner, Transformer, Aggregator, Update Strategy, Rank, Lookup, Filter, Sorter, Source Qualifier, Stored Procedure transformation, etc.**
- Extensive working experience in Apache Stack: **HDFS, Hadoop, Hive, MapReduce, Spark, Kafka, Zookeeper, Airflow, Flume, Super Set**
- Scheduled and Orchestrated data applications for both real-time event triggering jobs and batch in both **AWS and GCP Cloud**
 - ⇒ In AWS Cloud I have used services like: **Cloud Watch Event, Lambda, SNS Topic, Airflow, Step Function, S3, Glue (Jobs, Connections, Notebooks, Crawlers & Catalog), Redshift, Cloud Watch logs**
 - ⇒ In GCP Cloud I have used services like: **Cloud Scheduler, Workflow, Cloud Composer, Pub-Sub, DataFlow, BigQuery, Cloud SQL**
- Migrated legacy data applications from **Hadoop** ecosystem to **AWS Cloud and GCP Cloud**.
- Used **QlikView, Kibana, Tableau and Apache superset** for visualization.

PROFESSIONAL EXPERIENCE:

JP Morgan Chase & Co, Albany, NY

Sep 2021- Till Date

Role: Senior Data Engineer

Description: As a team we developed a Data Lake and Data Warehouse for Analytics. Also designed and developed Machine learning platform and Data pipelines to validate Machine learning models. Also developed an ETL pipeline using cutting edge technologies to see model performance on Dashboards for monthly, daily and hourly basis. Also developed analytical feature repository using AWS services, so machine learning engineers, Data Scientist and Data analysts can use those features, this has decreased significant amount of time they spent of recreating feature and instead focus on model building.

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Roles & Responsibilities:

- Involved with project team members to deliver data models to meet data requirements and create all documentation and deliverables in accordance with enterprise guidelines and standards. Work on Metadata transfer amongst various proprietary systems.
- Partnered with ML Engineers and Data Scientists to build AI-driven solutions including chatbots, RAG pipelines, and embedding workflows leveraging **AWS Gen AI** services such as **Amazon Bedrock**, **Amazon Titan models**, **Amazon SageMaker JumpStart**, and **SageMaker Pipelines**.
- Designed and implemented **Retrieval-Augmented Generation (RAG)** architecture using Amazon **Kendra** for semantic search and **Bedrock-hosted LLMs** for response generation.
- Designed and maintained scalable data pipelines supporting **Retrieval-Augmented Generation (RAG)** architecture to power intelligent conversational agents with real-time contextual knowledge.
- Developed chatbot frameworks integrated with **Amazon Lex**, **Comprehend**, and **Bedrock** to support intelligent, multilingual, and context-aware conversations.
- Created vector-based feature stores and embedding pipelines using **SageMaker** + FAISS, enabling efficient semantic search across structured and unstructured data.
- Designed and developed data pipeline for **Gen AI (Generative Artificial Intelligence) & machine learning platform**.
- Designed and developed data pipeline using **Docker**, **Kubernetes**, **Logstash**, **Hadoop** and **AWS Stack**.
- Automated daily **ETL** process using **Apache Airflow** and **Crontab**.
- Created **Terraform resources** and **modules** to deploy AWS services like **S3 buckets**, **Lambda Functions**, **Glue Jobs**, **EMR clusters**, **Step Functions**, **I AM roles and Policies**, **Athena** and **QuickSight**.
- Created **YML** files to use **Docker** images, configured to **GitLab CI/CD** pipeline with **Terraform**.
- Created a mechanism (Data pipeline) to collect data from application running in **Docker** and **Kubernetes** to store data in **Hadoop** and **AWS**.
- Moved large amount of data from **AWS S3** buckets to **AWS Redshift** using **Glue** and **EMR**.
- Analysed large and critical datasets using **EMR**, **Glue** and **Spark**.
- Experienced in writing live **Real-time Processing** and core jobs using **Spark Streaming** with **Kafka** as a data pipe-line system.
- Coordinated daily, weekly, bi-weekly, monthly, and quarterly oversight monitoring activities using **AWS QuickSight**.
- Developed ETL pipelines in and out of data warehouse using tools like **Python** and **AWS Glue**.
- Implemented Spark using **PySpark** and **Spark SQL** for faster testing and processing of data.
- Consumed the data from **Kafka** using **Apache spark**.
- Used **Spark** and **Spark-SQL** to read the parquet data and create the tables in hive using the **Scala API**.
- Building models in spark using python in **Databricks** and **SageMaker Notebooks**.

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- Worked on migrating **SQL** scripts from **Redshift** and **Athena**.
- Experience in building CI/CD pipelines for testing and production environments using **Terraform**.
- Proficient with container systems like **Docker** and container orchestration like **EC2 Container Service**, **Kubernetes**, worked with **Terraform**.
- Worked on POC's with **Apache Spark** using **Scala** to implement **spark** in project.
- Used **AWS EMR** to **transform** and move large amounts of data into and out of other **AWS** data stores and databases, such as **Amazon Simple Storage Service (Amazon S3)** and **Amazon DynamoDB**.
- Done some POC on new technologies. Followed agile methodology. Suggested and implemented new technologies.

RCG Atlanta, GA

Jan 2018 – Aug 2021

Role: Data Engineer

Description: As part Data Analytical Team, responsibility is to enable data sources (Data Lake and Data Warehouse) for analytical team

Roles & Responsibilities:

- Designed, developed and deployed **DataLake** to store Realtime and batch data from applications and databases using **Confluent Kafka** and **AWS S3**.
- Designed, developed and deployed **Datawarehouse** using **AWS Redshift** following best practices and optimized the queries performance and storing the tables.
- Used **AWS Fargate** and **ECS** for scalable deployment of containerized end points inference services built using **Fast API** and Python.
- Integrated **IAM**, **API Gateway**, and **CloudWatch** with **Machine Learning Models** to ensure secure, auditable, and highly available model access patterns.
- Developed self-service pipelines for internal data teams using **SNS**, **Lambda**, **Glue**, **EMR**, and **Athena**, enabling data readiness for **Machine Learning** and **Data Science** workloads.
- Designed, developed and deployed batch pipelines using **SNS**, **Lambda**, **StepFunction**, **Glue**, **SageMaker**, and **EMR**.
- Used **QlikView**, **Kibana**, **Tableau** and **Apache superset** for visualization.
- Build Self-service data pipelines using **AWS Services** like, **SNS**, **StepFunction**, **Lambda**, **Glue**, **EMR**, **EC2**, **Athena**, **SageMaker**, **QuickSight**, **Redshift** etc.
- Designed **database architecture** and developed self-service **data platform** in **AWS Cloud**.
- Designed and developed **data pipelines**. Automated based on the requirement.

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- Created **Terraform resources** and **modules** to deploy AWS services like **S3 buckets, Lambda Functions , Glue Jobs, EMR clusters , Step Functions, I AM roles and Policies, Athena and QuickSight.**
- Developed complex data pipelines using **AWS resource S3, EC2, Lambda, Glue, EMR, Step Function and RDS.**
- Developed Topic extraction (**Topic Modelling**) model using **LDA** and other customized **NLP** techniques.
- Developed and automated **data normalizing** model using Python libraries.
- Developed process to generate missing data from source files.
- Designed and developed **data cooking model** to normalize and transform raw data.
- Gathering requirements from SME, Project Manager and Business analysis.
- Established **AWS Bastion host** for secure data transaction between application and database team.
- Represented team in few summits.
- Created **POC on Maria DB Audit Plugin**, identified few performance issues then escalated to **Maria DB research and Develop team.**

S&P Global Charlottesville, VA

Feb 2016- Dec 2017

Role: Data Engineer

Description: Involving in **S&P Global** is prestige's 360-degree profiles assembling project. Analysing multi-dimensional, conducting Network optimization, Predictive analytics, and Social Analytics.

Roles & Responsibilities:

- Designed, developed and deployed **DataLake, DataMart, Datawarehouse** in **Google Cloud Platform.**
- Designed, developed and deployed batch and Realtime data pipelines using **GCP services.**
- Extensive experience on Google cloud services like **Google Cloud Storage, Cloud Function, DataFlow, BigQuery, Cloud Composer (Airflow), DataProc, Compute Engine, Cloud Run** etc.
- Designed and developed and deployed Realtime streaming pipelines using **Apache Kafka** and **Confluent Kafka** to ingest Realtime and batch data from **Relational Databases, DataLakes** and **IOT applications data.**
- Designed and developed and deployed various batch data pipelines using **Airflow.**
- Developed various dags, plugins and custom hooks in **Airflow.**
- Designed, developed and deployed streaming pipelines using **PubSub** and **Dataflow.**
- Migrated On-prem **Hadoop ecosystem** DataLake, Datawarehouse and Data Pipelines to **Google cloud Platform.**
- Created Realtime consumers to read data from **Kafka** and load data into **DataLake (Cloud Storage)** and transformed data using **Spark** in **DataProc** cluster.

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- Designed and developed streaming data pipelines using **Apache Kafka, Flume, Hive** and **Pig**.
- Built complex ETL process and stored process data in **Hive** and **HDFS**.
- Transformed raw text data from blogs and social networking sites provided from third party vendors for **sentimental analysis, information extraction** and **information retrieval**.
- Performed data profiling to learn the behaviour of various features such as traffic pattern, location, time, Date and Time etc.
- Good understanding on machine learning techniques like **supervised** and **unsupervised learning**.
- Developing database which contains **Tables, Stored Procedures, Functions, Views, Triggers** and **Indexes in SQL SERVER** and connecting to existing **CMS system**.
- Created a POC on **image classification model**.
- Understating sales from past few years, using **Machine learning techniques**.
- Extracted data from third party application and conducted data preprocessing and data mining using **R** and **Python**.

IBM, Hyd IN.

Jul 2014- Dec 2015

Role: Data Analyst

Description: As a Data Mining team, we are responsible to provide data-specific IT services across multiple domains, industries and verticals. Our organization has expanded its resources and knowledge base by focusing on the delivery of the very latest technologies and methodologies that successfully meet customers' IT challenges.

Roles & Responsibilities:

- Participated in analysis, design, development, testing, and implementation of various financial Systems using **Oracle, Developer** and **PL/SQL**.
- This system consists of the various functional modules.
- Define database structure, mapping and transformation logic. Creation of External Table scripts for loading the data from source for **ETL (Extracting Transforming and Loading) Jobs**.
- Wrote **UNIX Shell** Scripts to run database jobs on server side.
- Developed new and modified **existing packages, Database triggers, stored procedure** and **other code modules using PL/SQL** in support of business requirements.
- Worked with various functional experts to implement their functional knowledge into business rules in turn as working code modules like procedures and functions.

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TOOLS AND TECHNOLOGIES:

Languages	Shell, SQL, Python, PySpark and R
Generative AI & LLMs	LLMs: Amazon Titan, GPT-4 / GPT-3.5 (OpenAI), Claude (Anthropic), Gemini (Google), Llama 2 / 3 (Meta), Falcon, Mixtral, DeepSeek, Cohere, BLOOM
Embedding & Vector DBs	SentenceTransformers, Hugging Face Embeddings, FAISS, ChromaDB, Pinecone, Weaviate, Milvus, OpenSearch
RAG & Agentic Frameworks	LangChain, Llama Index, Haystack, CrewAI, AutoGen, PromptLayer
Packages	NumPy, Pandas, PyOd, Spacy, Matplotlib, Seaborn, Bokeh, Beautiful Soup, Selenium, Scikit-learn, TensorFlow, PyTorch, Lime, H2O, Scikit-image, Pillow, Psycopg, SQLAlchemy , Flask and more.
Container technologies	AWS EKS, Docker and Kubernetes.
Cloud Platforms	AWS and GCP.

EDUCATION:

Master of Computer Applications <i>Osmania University, Telangana, India</i>	Jun 2014
Bachelor of Science B.sc (Mathematics. Physics. Chemistry) <i>Kakatiya University, Telangana, India</i>	Jun 2010